# Logarithmic Weighted Modular Control for the Half-Filled Sine Kernel

Kirtus Justus (Theoretical Physicist) • 31 October 2026
Port Coquitlam, British Columbia, Canada

## Abstract

For the finite interval of a half-filled infinite discrete sine-kernel projection, we establish an explicit logarithmic upper bound on the occupation-weighted quadratic discrepancy between its exact modular generator and the established commuting-Jacobi scaling function. The proof uses an elementary trace-class estimate for the half-line sine-kernel Hankel blocks and a scalar logarithmic inequality. Combined with the independently established Wilson–Dirac positive-majorant pairing estimate, this shows that the *transverse* contribution to the fixed-scaled-mass relative-entropy modular correction is sublogarithmic. An elementary fractional spectral-moment bound yields a considerably weaker sufficient condition for the leading logarithmic law than uniform boundedness of the modular-error operator. The separate argument in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law* supplies the longitudinal bound and leading entropy law. Neither the present estimate nor that leading law determines the finite part or its proposed inverse-logarithmic remainder rate.

## 1. Reference operators and established inputs

Let $N = 2L \geq 2$. On $\mathbb{C}^N$ define the half-filled sine-kernel covariance

$$(B_N)_{jk} = \begin{cases} \frac{1}{2}, & j = k, \\ \dfrac{\sin(\pi(j-k)/2)}{\pi(j-k)}, & j \neq k, \end{cases} \qquad 0 \leq j, k < N.$$

Let $J_N$ be the real-symmetric tridiagonal matrix with zero diagonal and $(J_N)_{j-1,j} = j(N-j)/N$ for $1 \leq j < N$. These matrices commute, $0 < B_N < I$, and $\|J_N/N\| \leq 1/2$. The strict occupation bounds follow from the fact that a nonzero finitely supported sequence has Fourier transform that cannot vanish identically on either of the two nonempty half-circle frequency arcs.

Set

$$W_N = B_N(I - B_N), \quad K_N = \log(I - B_N) - \log B_N,$$
$$\eta(t) = \pi t \, {}_3F_2\left(\tfrac{1}{4}, \tfrac{1}{2}, \tfrac{3}{4}; 1, \tfrac{3}{2}; 4t^2\right), \quad E_N = K_N + N\eta(J_N/N).$$

The generalized hypergeometric series has nonnegative coefficients and converges at its argument 1. Therefore

$$A_\eta := \pi \, {}_3F_2\left(\tfrac{1}{4}, \tfrac{1}{2}, \tfrac{3}{4}; 1, \tfrac{3}{2}; 1\right) < \infty, \qquad |N\eta(\lambda/N)| \leq A_\eta|\lambda|$$

for every $|\lambda| \leq N/2$. We use the separately proved reference Jacobi moment estimate

$$M_{2,N} = \mathrm{Tr}(J_N^2 W_N) \leq \frac{32}{\pi^2}(H_N + 1), \quad H_N = \sum_{k=1}^{N} \frac{1}{k}. \tag{1}$$

The sharper identity $M_{2,N} = (3\pi^2)^{-1} \log N + O(1)$ is available, but unnecessary below.

## 2. Explicit trace-class estimate for the reference crossing

Let $P$ be the orthogonal projection on $\ell^2(\mathbb{Z})$ onto the Fourier arc $(-\pi/2, \pi/2)$, and let $\chi_N$ project onto $N$ consecutive sites. Then $B_N = \chi_N P \chi_N$, as an operator on the interval. Define $X_N = \chi_N P(I - \chi_N)$. The full-line projection identity yields

$$X_N X_N^* = B_N(I - B_N) = W_N, \qquad \operatorname{Tr} \sqrt{W_N} = \|X_N\|_1. \tag{2}$$

For either exterior half-line, after reversing the interior index, the cross-boundary matrix has entries

$$\frac{\sin\{\pi(i+j+1)/2\}}{\pi(i+j+1)}, \qquad 0 \le i < N, \quad j \ge 0.$$

Write $H^{(0)}_{ij} = 1/(i+j+1)$. The sine factor is a difference of two complex exponentials; each gives diagonal unitary modulation of $H^{(0)}$. Therefore the nuclear norm of each half-line block is at most $\pi^{-1}\|H^{(0)}\|_1$.

The elementary Laplace representation

$$H^{(0)}_{ij} = \int_0^\infty e^{-t(i+j+1)}\, dt$$

is a rank-one Bochner integral. Its integrand has nuclear norm

$$\mathcal{W}_N(t) = e^{-t} \frac{\sqrt{1 - e^{-2Nt}}}{1 - e^{-2t}}.$$

Consequently, for the union of both exterior half-lines,

$$\operatorname{Tr} \sqrt{W_N} = \|X_N\|_1 \le \frac{2}{\pi} I_N, \qquad I_N = \int_0^\infty \mathcal{W}_N(t)\, dt. \tag{3}$$

For $0 < t \le 1$, the concavity inequality $1 - e^{-2t} \ge (1 - e^{-2})t$ and $1 - e^{-2Nt} \le \min\{2Nt, 1\}$ give

$$\mathcal{W}_N(t) \le \frac{1}{1 - e^{-2}} \min\left\{ \sqrt{\frac{2N}{t}}, \frac{1}{t} \right\}.$$

For $t \ge 1$, $\mathcal{W}_N(t) \le e^{-t}/(1 - e^{-2})$. Splitting the integral at $t = 1/N$ and $t = 1$ gives

$$\boxed{I_N \le \frac{\log N + 2\sqrt{2} + e^{-1}}{1 - e^{-2}}.} \tag{4}$$

All constants are independent of $N$. This is a deliberately nonoptimal estimate.

## 3. Uniform quadratic modular estimate

**Theorem 1.** For every even $N \geq 2$,

$$\boxed{\operatorname{Tr}(W_N E_N^2) \leq \frac{16}{\pi} I_N + 2A_\eta^2 M_{2,N} \leq \frac{16(\log N + 2\sqrt{2} + e^{-1})}{\pi(1-e^{-2})} + \frac{64A_\eta^2}{\pi^2}(H_N + 1).} \tag{5}$$

In particular, $\operatorname{Tr}(W_N E_N^2) = O(\log N)$.

*Proof.* For $b \in (0,1)$ let $k = \log((1-b)/b)$ and $s = \sqrt{b(1-b)} = 1/(2\cosh(k/2))$. Since $\cosh(k/2) \geq k^2/8$,

$$b(1-b)k^2 = s^2k^2 \leq 4s. \tag{6}$$

Functional calculus gives $\operatorname{Tr}(W_N K_N^2) \leq 4 \operatorname{Tr} \sqrt{W_N}$. Because $B_N$, $J_N$, $K_N$, $E_N$ and $W_N$ commute,

$$\operatorname{Tr}(W_N E_N^2) \leq 2 \operatorname{Tr}(W_N K_N^2) + 2 \operatorname{Tr}\{W_N [N\eta(J_N/N)]^2\} \leq 8 \operatorname{Tr} \sqrt{W_N} + 2A_\eta^2 M_{2,N}.$$

Substitute (1), (3), and (4). □

**Qualification.** The estimate (5) is far weaker than the numerically observed decay of the weighted variance and does not itself prove a decaying variance bound. The stronger estimate $O(\log N/N^2)$ is established in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law*. Its value is that it is independent of any unproved pointwise modular-eigenvalue comparison.

## 4. Consequence for the Wilson–Dirac modular correction

For the mass-scaled Wilson–Dirac covariance perturbation $\Delta_N$, use its established full-line-projector decomposition compressed to the interval, $\Delta_N = S_N + D_N$, with positive operator $G_N$. In the common eigenbasis $v_j$ of $B_N, J_N$, put $b_j = \langle v_j, B_N v_j\rangle$, $\lambda_j = \langle v_j, J_N v_j\rangle$, $q_j = \langle v_j, G_N v_j\rangle$, $e_j = \langle v_j, E_N v_j\rangle$. Established inputs are

$$\operatorname{Tr} G_N \leq M_\mu := \frac{\mu}{\pi}, \quad |\langle v_j, S_N v_j\rangle| \leq q_j, \quad |\langle v_j, D_N v_j\rangle| \leq 2\sqrt{q_j b_j (1-b_j)}.$$

**Corollary 2 (transverse modular correction).** For each fixed $\mu > 0$,

$$\boxed{|\operatorname{Tr}(D_N E_N)| \leq 2\sqrt{\frac{\mu}{\pi} \operatorname{Tr}(W_N E_N^2)} = O_\mu(\sqrt{\log N}) = o(\log N).} \tag{7}$$

*Proof.* Apply the pairing bound, sum over common eigenvectors and use Cauchy–Schwarz. Then apply Theorem 1. □

Combining (7) with the exact two-bond asymptotic established in the companion Wilson reference-geometry result, bounded nonlinear correction via the Jacobi fourth moment, and bounded Wilson cross-boundary entropy difference yields the leading-order reduction

$$\boxed{\mathcal{D}_L(\mu) = \frac{\mu^2}{6} \log L + \operatorname{Tr}(S_N E_N) + O_\mu(\sqrt{\log L}), \qquad N = 2L.} \tag{8}$$

The leading coefficient follows if $\operatorname{Tr}(S_N E_N) = o(\log L)$. Neither (7) nor (8) proves this condition alone; *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law* supplies the stronger estimate $O_\mu(1)$.

## 5. Fractional spectral moments and weaker sufficient hypotheses

The established positive-majorant spectral-tail estimate is

$$\nu_N((t,\infty)) := \sum_{|\lambda_j|>t} q_j \le \min\left\{M_\mu, \frac{a_\mu}{t} + \frac{b_\mu}{t^2}\right\}, \quad a_\mu = \sqrt{\frac{2}{15}}\frac{\mu^2}{\pi}, \quad b_\mu = \frac{\mu}{3\pi}. \tag{9}$$

**Lemma 3 (uniform fractional majorant moment).** For every $0 < \alpha < 1$ and every even $N$,

$$\boxed{\sum_j q_j |\lambda_j|^\alpha \le M_\mu + \frac{\alpha a_\mu}{1-\alpha} + \frac{\alpha b_\mu}{2-\alpha}.} \tag{10}$$

*Proof.* Use layer-cake: $\sum q_j|\lambda_j|^\alpha = \alpha \int_0^\infty t^{\alpha-1}\nu_N((t,\infty))dt$. Integrate the total-mass bound on $(0,1)$ and the inverse-power bound on $(1,\infty)$. □

**Corollary 4 (sufficient sublinear spectral envelope).** If there exist $0 < \alpha < 1$ and $C_\alpha < \infty$, independent of even $N$, such that

$$|e_j| \le C_\alpha(1 + |\lambda_j|^\alpha) \quad \text{for every reference eigenpair,} \tag{11}$$

then $|\operatorname{Tr}(S_N E_N)| = O_{\mu,\alpha}(1)$, and

$$\mathcal{D}_L(\mu) = \frac{\mu^2}{6}\log L + O_\mu(\sqrt{\log L}). \tag{12}$$

The envelope (11) is only a **sufficient, unresolved hypothesis**, and is weaker at large $|\lambda|$ than a logarithmic eigenpair envelope.

**Alternative sufficient condition (uniform sublinearity).** One can weaken (11) further: suppose that for each fixed $R < \infty$, the modular errors are uniformly bounded on $|\lambda_j| \le R$, and

$$\lim_{R\to\infty} \sup_{N \text{ even}, j:\,|\lambda_j|>R} \frac{|e_j|}{|\lambda_j|} = 0. \tag{13}$$

Then $\operatorname{Tr}(S_N E_N) = o(\log N)$ and the leading logarithmic law follows. Indeed, for any $\varepsilon > 0$, (13) bounds $|e_j| \le \varepsilon|\lambda_j|$ outside a fixed window while the window contributes $O(1)$. By layer cake and (9), $\sum_j q_j|\lambda_j| \le M_\mu + a_\mu \log N + b_\mu$, up to harmless fixed-size adjustments. Dividing the resulting estimate by $\log N$, taking the limsup, then letting $\varepsilon \downarrow 0$, proves the claim. Condition (13) is **not currently established**.

## 6. Scope and sources

Theorem 1, Corollary 2, and Lemma 3 are analytic consequences of the stated finite sine-kernel model and the referenced, separately established Jacobi/majorant results. They use no numerically inferred uniform limit. The longitudinal modular estimate is supplied separately in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law*. None of these statements proves the proposed full finite part or the $O_\mu((\log L)^{-1/103})$ remainder.

Classical references: V. Eisler and I. Peschel, *Analytical Results for the Entanglement Hamiltonian of a Free-Fermion Chain*, J. Phys. A **50** (2017), 284003, doi:10.1088/1751-8121/aa76b5; D. Slepian, *Prolate Spheroidal Wave Functions, Fourier Analysis, and Uncertainty, V: The Discrete Case*, Bell Syst. Tech. J. **57** (1978), 1371–1430; V. Peller, *Hankel Operators and Their Applications*, Springer (2003), doi:10.1007/978-0-387-21681-2. These works furnish established mathematical context; they do not externally verify the Wilson-specific results presented here.

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